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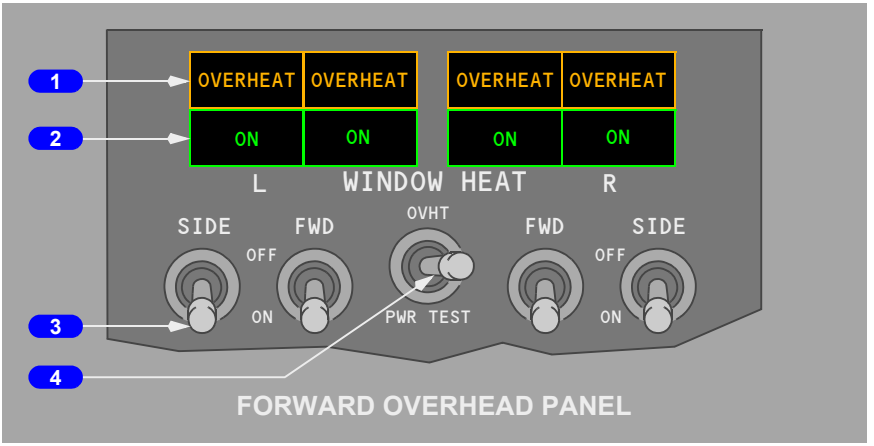
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Window Heat Panel



1 Window OVERHEAT Lights

Illuminated (amber) – overheat condition is detected.

Note: OVERHEAT lights also illuminate if electrical power to window(s) is interrupted.

2 Window Heat ON Lights

Illuminated (green) – window heat is being applied to selected window(s).

Extinguished –

- switch is OFF, or
- an overheat is detected, or
- a system failure has occurred
- system is at correct temperature.

3 WINDOW HEAT Switches

ON – window heat is applied to selected window(s).

OFF – window heat not in use.

4 WINDOW HEAT Test Switch (spring–loaded to neutral)

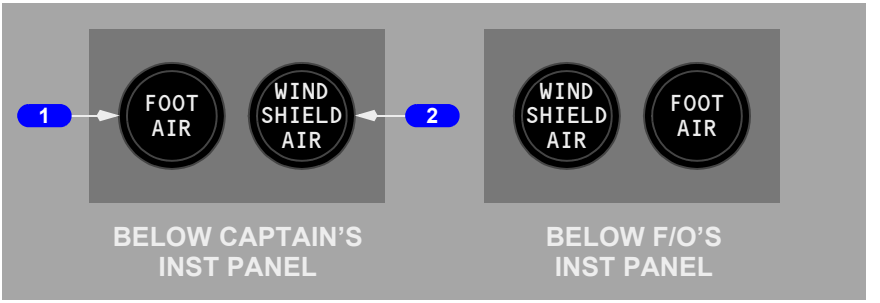
OVHT – simulates an overheat condition.

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PWR TEST – provides a confidence test.

Note: Refer to Supplementary Normal Procedures for Window Heat Test procedures.

Windshield/Foot Air Controls



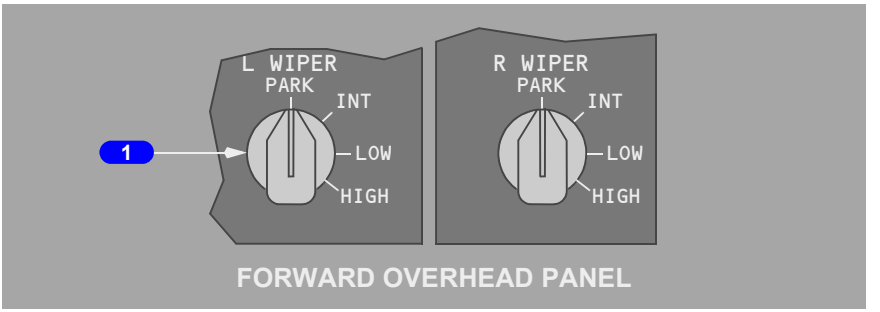
1 FOOT AIR Controls

PULL – supplies conditioned air to pilots' leg positions.

2 WINDSHIELD AIR Controls

PULL – supplies conditioned air to number 1 windows for defogging.

Windshield Wiper Selector Panel



1 Windshield WIPER Selectors

PARK – turns off wiper motors and stows wiper blades.

INT – seven second intermittent operation.

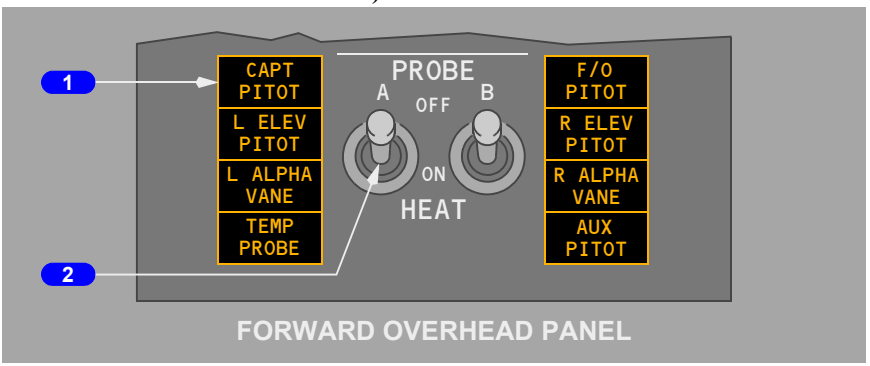
LOW – low speed operation.

HIGH – high speed operation.

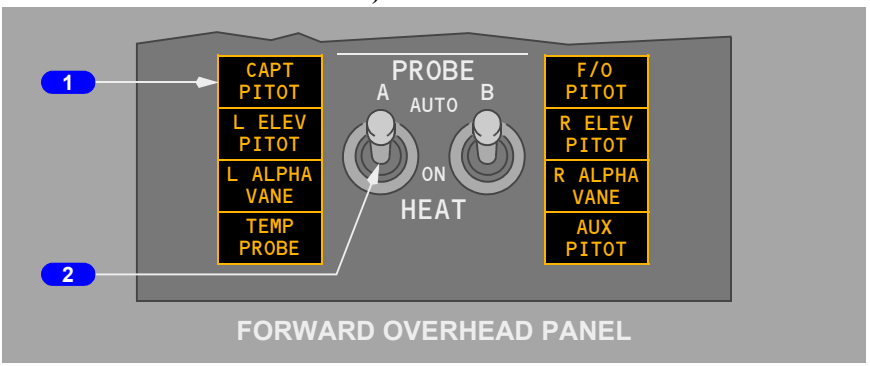
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Probe Heat Panel

YF592
(SB Changes YA573 ; before SB replaces OFF-ON Probe Heat switches with AUTO-ON Probe Heat switches.)



YC496 through YF591, YF593 through YW164
(SB Changes YA573 ; after SB replaces OFF-ON Probe Heat switches with AUTO-ON Probe Heat switches.)



1 Probe Heat Lights

Illuminated (amber) – related probe not heated.

Note: If operating on standby power, probe heat lights do not indicate system status.

YF592
(SB Changes YA573 ; before SB replaces OFF-ON Probe Heat switches with AUTO-ON Probe Heat switches.)

2 PROBE HEAT Switches

ON – power is supplied to heat related system.

OFF – power off.

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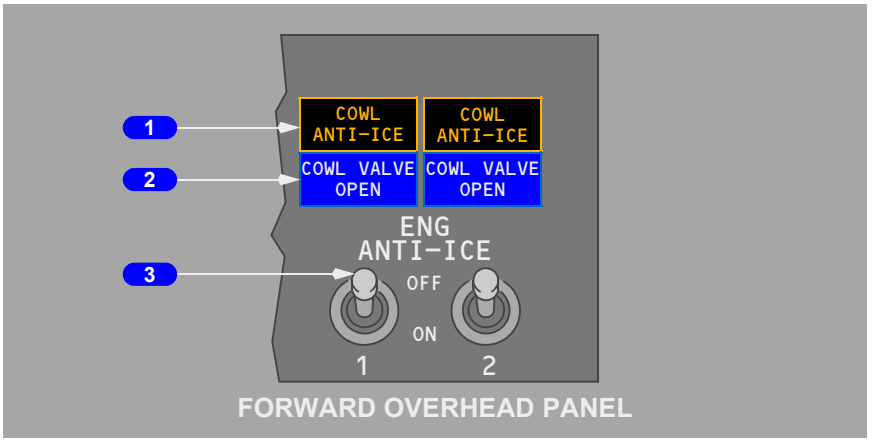
YC496 through YF591, YF593 through YW164
(SB Changes YA573 ; after SB replaces OFF-ON Probe Heat switches with AUTO-ON Probe Heat switches.)

2 PROBE HEAT Switches

ON – power is supplied to heat related system.

AUTO – power is automatically supplied to both A and B probe heat systems when either engine is running.

Engine Anti–Ice Panel



1 COWL ANTI–ICE Lights

Illuminated (amber) – indicates an overpressure condition in duct downstream of engine cowl anti–ice valve.

2 COWL VALVE OPEN Lights

Illuminated (blue) –

- bright – related cowl anti–ice valve is in transit, or, cowl anti–ice valve position disagrees with related ENGINE ANTI–ICE switch position
- dim – related cowl anti–ice valve is open (switch ON).

Extinguished – related cowl anti–ice valve is closed (switch OFF).

3 ENGINE ANTI–ICE Switches

ON –

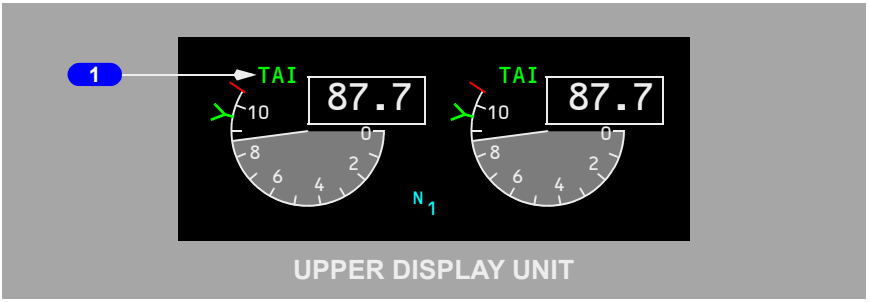
- related cowl anti–ice valve is open
- stick shaker logic is set for icing conditions.

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OFF –

- related cowl anti–ice valve is closed
- stick shaker logic returns to normal if wing anti–ice has not been used in flight.

Thermal Anti–Ice Indication

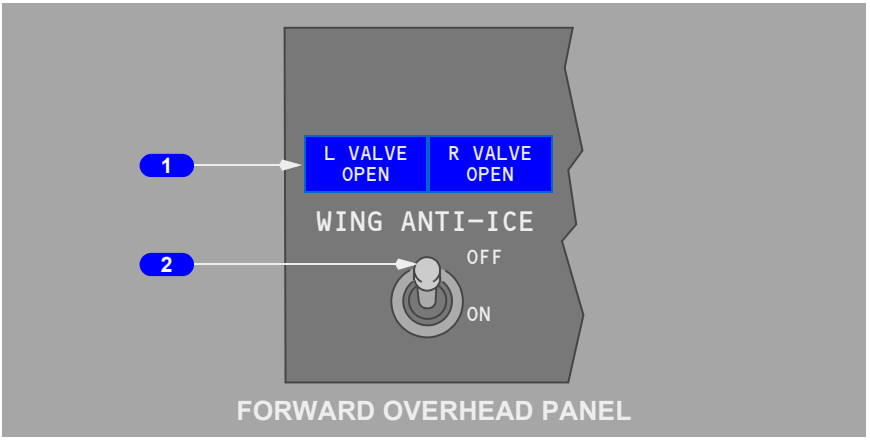


1 Thermal Anti–Ice Indications

Illuminated –

- Green – cowl anti–ice valve(s) open
- Amber – cowl anti–ice valve is not in position indicated by related engine anti–ice switch.

Wing Anti–Ice Panel



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1 Wing Anti–Ice VALVE OPEN Lights

Illuminated (blue) –

- bright – related wing anti–ice control valve is in transit, or, related wing anti–ice control valve position disagrees with WING ANTI–ICE switch position
- dim – related wing anti–ice control valve is open (switch ON).

Extinguished – related wing anti–ice control valve is closed (switch OFF).

2 WING ANTI–ICE Switch

OFF – wing anti–ice control valves are closed.

ON (in the air) –

- wing anti–ice control valves are open
- stick shaker logic is set for icing conditions.

Note: Stick shaker logic remains set for icing conditions for the remainder of the flight, regardless of subsequent WING ANTI–ICE switch position.

ON (on the ground) –

- wing anti–ice control valves open if thrust on both engines is below takeoff warning setting and temperature inside both distribution ducts is below thermal switch activation temperature
- control valves close if either engine thrust is above takeoff warning setting or thermal switch is activated in either distribution duct. Switch remains ON
- switch trips OFF at lift–off.

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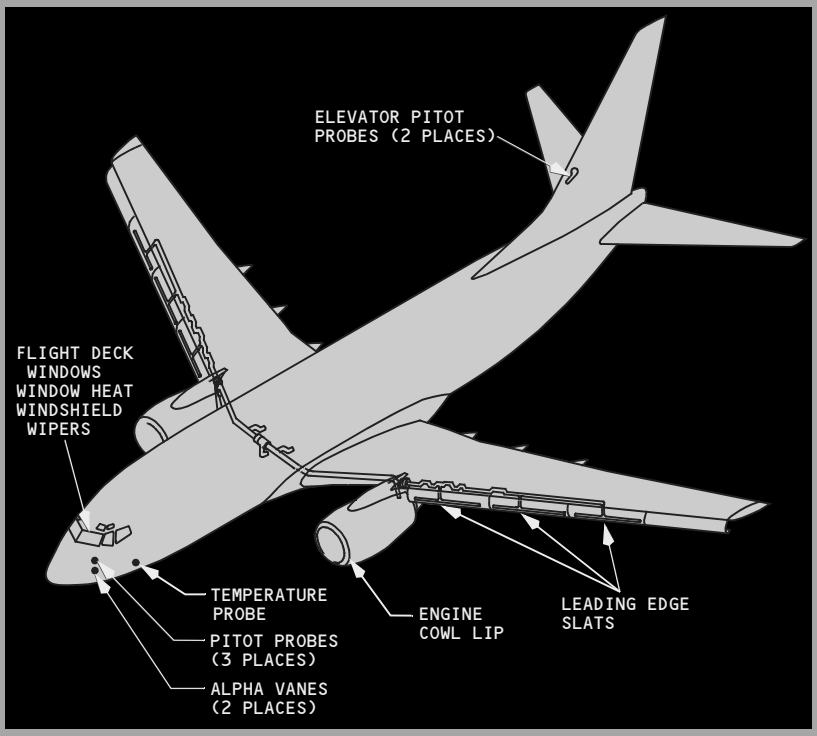
Introduction

Thermal anti-icing (TAI), electrical anti-icing, and windshield wipers are the systems provided for ice and rain protection.

The anti-ice and rain systems include:

- Flight Deck Window Heat
- Windshield Wipers
- Probe and Sensor Heat
- Engine Anti-Ice System
- Wing Anti-Ice System

Anti-Ice Components Diagram



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Flight Deck Window Heat

Flight deck window numbers 1 and 2 consist of glass panes laminated to each side of a vinyl core. Flight deck window number 3 consists of two acrylic panes separated by an air space.

A conductive coating on the outer glass pane of window numbers 1 and 2 permits electrical heating to prevent ice build–up and fogging. Window number 3 is not electrically heated.

Flight Deck Window Heat Operation

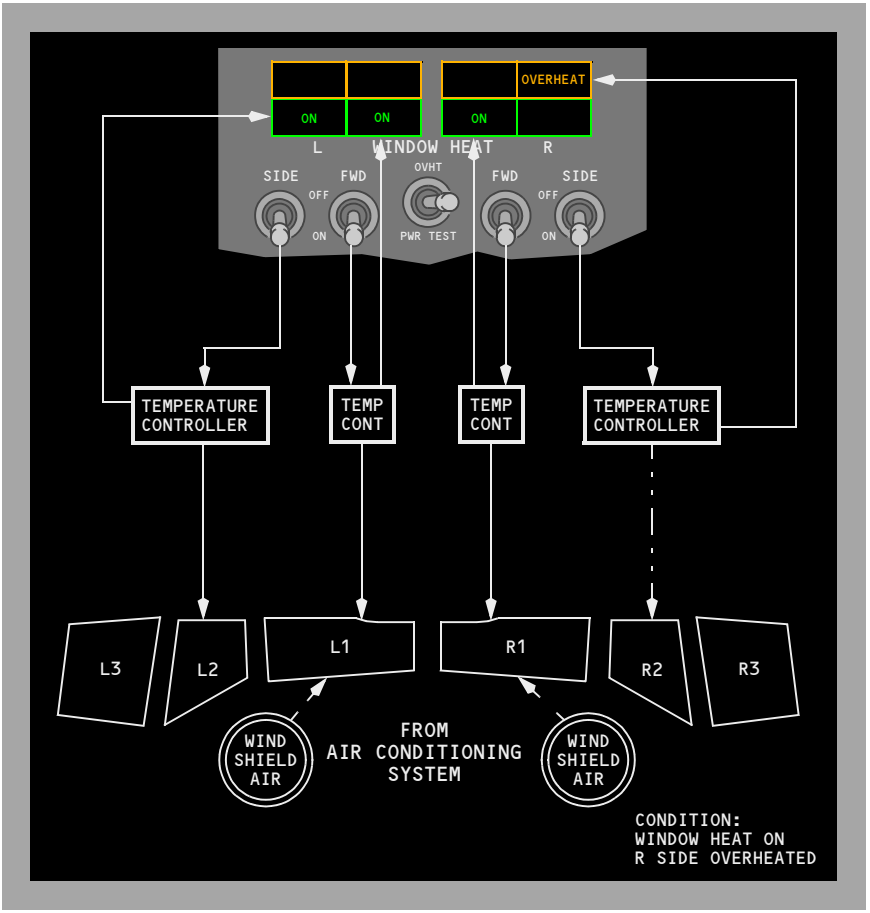
The FWD WINDOW HEAT switches control heat to window No. 1. The SIDE WINDOW HEAT switches control heat to window numbers 2.

Temperature controllers maintain window numbers 1 and 2 at the correct temperature to ensure maximum strength of the windows in the event of bird impact. Power to window numbers 1 and 2 is automatically removed if an overheat condition is detected.

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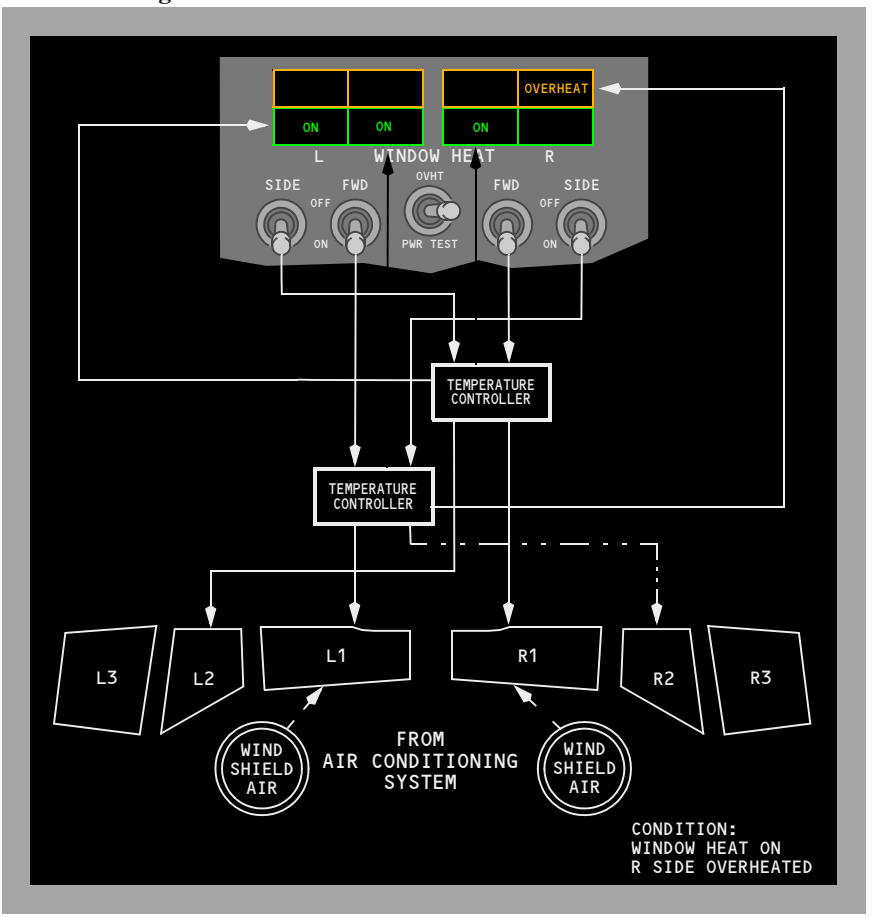
Flight Deck Window Heat Schematic

YA573 through YW028



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YW029 through YW164



Windshield Wipers

The rain removal system for the forward windows consists of windshield wipers and a permanent rain repellent coating on the windows.

CAUTION: Windshield scratching will occur if the windshield wipers are operated on a dry windshield.

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Probe and Sensor Heat

Pitot probes, the total air temperature probe and the alpha vanes are electrically heated. Static ports are not heated. When operating on standby power, only the captain's pitot probe is heated, however, the CAPT PITOT light does not illuminate for a failure.

Note: The pitot probe for standby airspeed is not heated when the airplane is on standby power.

Engine Anti–Ice System

Engine bleed air thermal anti–icing prevents the formation of ice on the engine cowl lip. Engine anti–ice operation is controlled by individual ENG ANTI–ICE switches. The engine anti–ice system may be operated on the ground and in flight.

Engine Anti–Ice System Operation

Each cowl anti–ice valve is electrically controlled and pressure actuated. Positioning the ENG ANTI–ICE switches to ON:

- allows engine bleed air to flow through the cowl anti–ice valve for cowl lip anti–icing
- sets stall warning logic for icing conditions.

Note: Stall warning logic adjusts stick shaker and minimum maneuver speed bars on the airspeed indicator. FMC displayed VREF is not adjusted automatically.

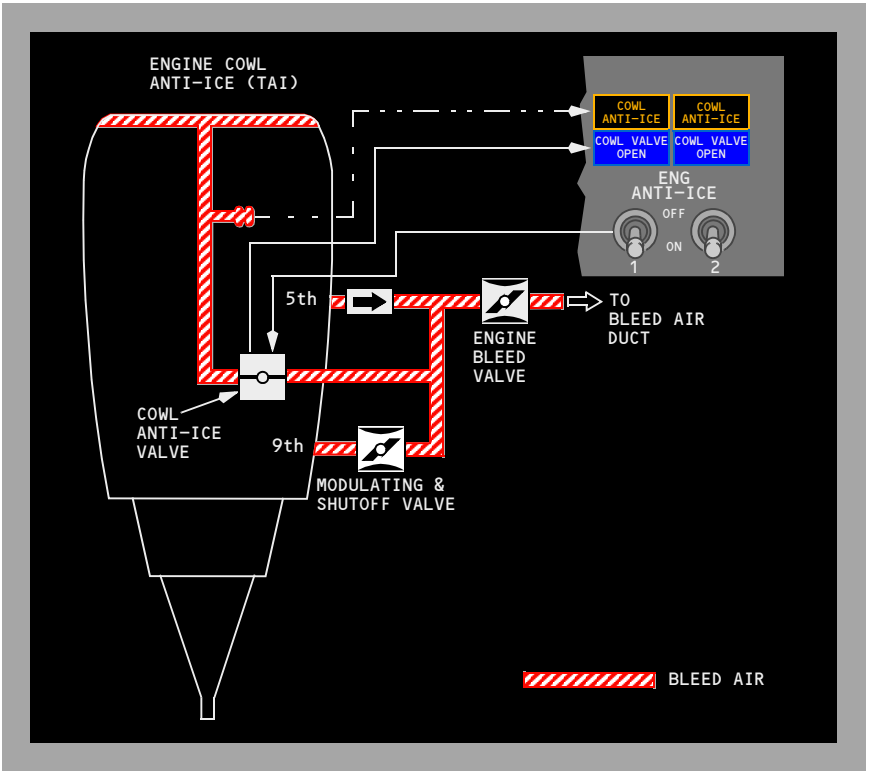
Note: Stall warning logic, airspeed indications, and minimum maneuver speeds on the airspeed indicator return to normal when engine anti–ice is positioned OFF if wing anti–ice has not been used in flight.

If the cowl anti–ice valve fails to move to the position indicated by the ENG ANTI–ICE switch, the COWL VALVE OPEN light remains illuminated bright blue and an amber TAI indication illuminates on the CDS after a short delay.

The amber COWL ANTI–ICE light illuminates due to excessive pressure in the duct leading from the cowl anti–ice valve to the cowl lip.

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Engine Anti–Ice System Schematic



Wing Anti–Ice System

The wing anti–ice system provides protection for the three inboard leading edge slats by using bleed air. The wing anti–ice system does not include the leading edge flaps or the outboard leading edge slats.

The wing anti–ice control valves are AC motor–operated. With a valve open, bleed air flows to the three leading edge inboard slats, and is then exhausted overboard. The wing anti–ice system is effective with the slats in any position.

Wing Anti–Ice System Operation

On the ground, positioning the WING ANTI–ICE switch ON opens both control valves if thrust on both engines is below the setting for takeoff warning activation and the temperature inside both wing distribution ducts is less than the thermal switch activation temperature.

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Both valves close if either engine thrust is above the takeoff warning setting or either temperature sensor senses a duct overtemperature. The valves automatically reopen if thrust on both engines is reduced and both temperature sensors are cool.

With the air/ground sensor in the ground mode and the WING ANTI–ICE switch ON, the switch remains in the ON position regardless of control valve position. The WING ANTI–ICE switch automatically trips OFF at lift–off when the air/ground sensor goes to the air mode.

Positioning the WING ANTI–ICE switch to ON in flight:

- opens both control valves
- sets stall warning logic for icing conditions.

Note: Stall warning logic adjusts stick shaker and minimum maneuver speed bars on airspeed indications. FMC displayed VREF is not adjusted automatically.

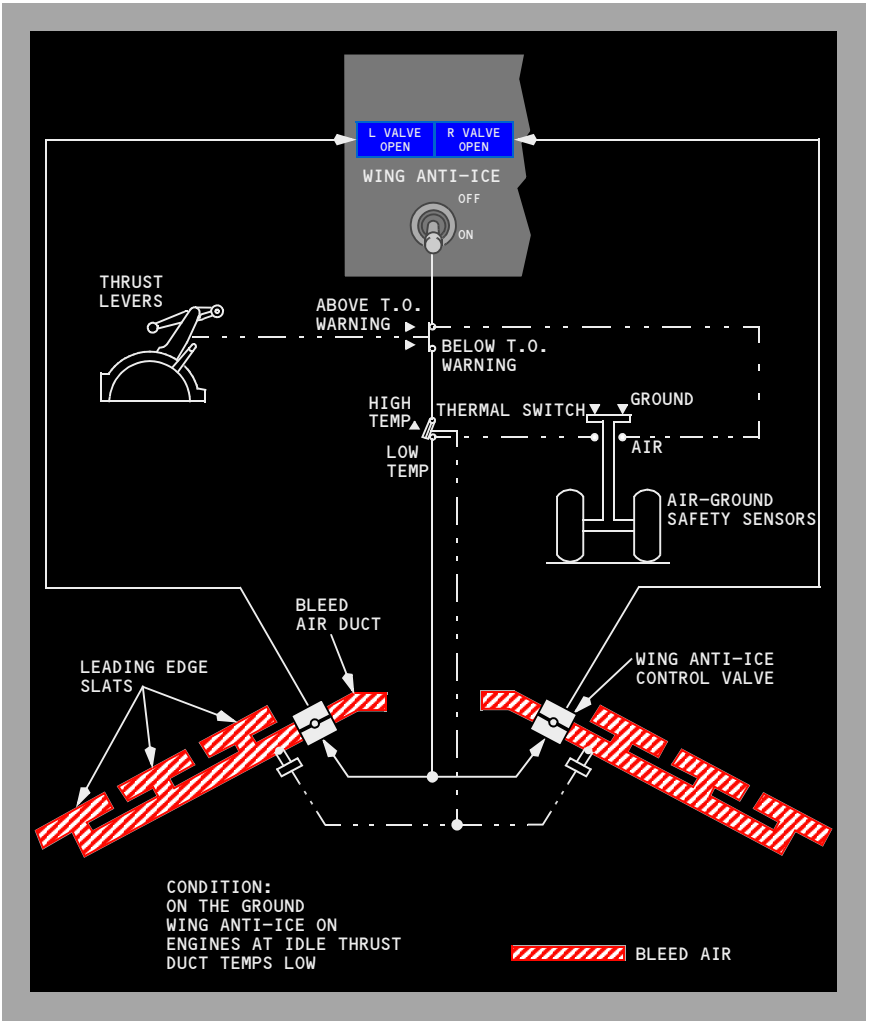
Note: Stall warning logic remains set for icing conditions for the remainder of the flight, regardless of subsequent WING ANTI–ICE switch position.

Duct temperature and thrust setting logic are disabled and have no effect on control valve operation in flight.

Valve position is monitored by the blue VALVE OPEN lights.

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Wing Anti–Ice System Schematic



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